

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Mechanical (CAD/CAM) Examination

December 2012

ME 704.01 Advanced Machine Design

Date: 19.12.2012, Wednesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. PSG Design data book will be provided if required.

SECTION - I

- Q - 1 (a) Discuss on design for rigidity. How rigidity of mechanical system and components can be increased. [03]
- Q - 1 (b) What is the difference between specification steps and defining problem steps in design process? [06]
- Q - 1 (c) What is the importance of various theories of failures in the design of mechanical components? Explain the von mises criteria and its application in engineering design. [06]
- Q - 2 (a) Explain miner's rule. [05]
- Q - 2 (b) Discuss on endurance strength and effect of modifying factors on endurance strength of materials. [05]

OR

- Q - 2 An alloy steel shaft made from material which has $\sigma_{ut}=700\text{MPa}$ and its endurance limit is $0.5*\sigma_{ut}$. It is subjected to a fluctuating stresses as shown: [10]

σ_m (MPa)	σ_a (MPa)	% time
400	300	20
350	250	40
450	500	40

Estimate the life of shaft. Assume suitable factors.

- Q - 3 An edge crack beam (Depth of the beam $w=50\text{mm}$, Thickness $b=20\text{mm}$, Span $S=200\text{mm}$), has a crack in its central plane whose length is 10mm . If two loads of 2000N each are placed at 50mm from central plane, what will be SIF of crack? [10]

OR

- Q - 3 (a) Enumerate the factors responsible for brittle fracture of materials, which normally in laboratory would behave as ductile materials. [04]
- Q - 3 (b) Derive the equations for bending stress in edged cracked three point and four point bend configurations. [06]

SECTION - II

- Q - 4 (a) Explain importance of concurrent engineering in QFD of a product design. [03]
- Q - 4 (b) Discuss human factor consideration in design and applications in brief. [06]
- Q - 4 (c) Boron fibers (40 v %) are used to reinforce aluminum for an engineering application. Use the table provided below to calculate the density, modulus of elasticity, and tensile strength long the fiber orientation. Also estimate the modulus of elasticity if the load is to be applied perpendicular to the fibers. [06]

Materials	Density (g/cm ³)	Modulus (GPa)	Strength (MPa)
Boron Fiber	2.36	55	4000
Aluminum	2.70	10	50

- Q - 5 Answer the following (*Any Two*). [08]
- (a) Explain in brief the system concept for material handling.
- (b) What are the basic objectives of a material handling system?
- (c) What do you understand by containerization?

OR

- Q - 5 Determine the belt capacity of a horizontal belt, if the effective width of the material on the belt is 'b', belt velocity is 'v' and the surcharge angle is 15°. [08]
- Q - 6 (a) Explain PDE and SDE in design optimization with necessary examples. [03]
- Q - 6 (b) Design a tensile bar (Square C/S) of length L= 300mm to carry a tensile load of 6 kN for minimum cost of the following materials: [09]

Material	Mass Density(kg/m ³)	Material Cost (Rs/N Weight)	Yield Strength (MPa)
Steel	7500	16	130
Al-alloy	3000	32	50
Titanium Alloy	4800	480	90
Magnesium Alloy	2100	32	20

OR

- Q - 6 (b) For S-Gearing determine the dimensions of gear keeping ration in center distance constant $m=7$, $\alpha=20^\circ$, $i=8.5$ and $a=182$ mm. Determine the diameter and contact ratio. [09]
